

Vaibhav Katariya

+91-7015000239 | vaibhav@kaily.in | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

Jaypee Institute of Technology

B.Tech. in Electronics and Communication Engineering

Noida, India

2023 – 2027

Air Force Bal Bharati School

Senior Secondary Education, Science Stream

New Delhi, India

2010 – 2023

Technical Skills

Programming Languages: JavaScript, Python, TypeScript, PHP, C/C++

Tech Stack: React, Next.js, Node.js, PostgreSQL, MongoDB, Tailwind CSS, WebSockets, Express

DevOps & Tools: Git, GitHub, AWS (Basic), Nginx, Reverse Proxies, Cloudflare, Rate Limiting, DDoS Protection

Projects

Autamail | *Next.js, Firebase Auth, Firestore, ZeptoMail, Nodemailer*

- Built a production-grade email automation platform used by the club during BitBox for sponsor and participant outreach, processing **1000+ emails** with zero duplication.
- Enabled bulk campaigns via CSV ingestion and **SHA-256 deduplication**, contributing to **Rs.1,00,000+ sponsorship outreach**.
- Implemented admin approval workflows and **RBAC** to control campaign execution and prevent unauthorized sends.
- Handled failure scenarios with retry queues and idempotent processing to ensure reliable delivery.
- Reduced **Firestore cost by 70%** using server-side APIs, pagination, and caching.

DENTOVAULT | *Next.js, Node.js, Express, PostgreSQL, Nginx, JWT*

- Architected and delivered an end-to-end medical data platform for a dental practitioner, including web portal, backend APIs, authentication, and storage systems.
- Designed a **local-first architecture** (no cloud) to meet strict privacy and low-latency requirements.
- Engineered hybrid storage (PostgreSQL + filesystem) managing **1800+ high-resolution patient images**.
- Implemented **JWT-based RBAC** and secure APIs to control access to sensitive medical data.
- Deployed on a local mini PC with Nginx and PM2, ensuring reliable operation within a private LAN.

Achievements

Winner - Ride Hack 3.0

Dec 2025

Sahyatri – AI-Based Public Transit Occupancy System

- Deployed an edge AI system on Raspberry Pi 4 under constrained compute conditions.
- Built a hybrid YOLOv5-based model achieving **99% accuracy** for passenger detection across varying lighting conditions.
- Developed a React dashboard for live occupancy tracking, integrating with FastAPI-based backend services.

Position of Responsibility

GDG — Google Developer Groups

Aug 2025 – Present

Technical Lead

- Led development of the club's **official website (gdg-jiit.com)** and organized BitBox 6.0, driving **2,500+ registrations**.